Chapter 10 — Electrochemistry

Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

1. Oxidation is: (a) Gain of electrons (b) Loss of electrons (c) Loss of neutrons (d) Gain of protons

2. Reduction occurs at: (a) Wire (b) Cathode (c) Anode (d) Salt bridge

3. Electrolytic cell converts: (a) Light to heat (b) Heat to light (c) Chemical to electrical (d) Electrical to chemical

4. Molten NaCl cathode product: (a) Na (b) O_2 (c) Cl_2 (d) H_2

5. SHE potential: (a) +1.00 V (b) +0.34 V (c) 0.00 V (d) -0.76 V

6. Cu_2+/Cu E°: (a) 0.00 V (b) +0.34 V (c) -0.76 V (d) -2.38 V

7. In Cu-Zn cell, anode: (a) Cu (b) Zn (c) H_2 (d) Salt bridge

8. Salt bridge maintains: (a) Electrical neutrality (b) Pressure (c) Temperature (d) Colour

Q.2 — SHORT QUESTIONS (6 × 2 = 12)

1. Why is oxidation always accompanied by reduction?

2. How does oxidation number identify the oxidizing agent?

3. Why does molten NaCl give sodium but aqueous NaCl gives hydrogen?

4. Why is the standard hydrogen electrode assigned zero potential?

5. Why is platinum used in the standard hydrogen electrode?

6. Why is a salt bridge needed in a galvanic cell?

Q.3 — LONG QUESTIONS (2 × 8 = 16)

Q.3 (a) Explain redox reactions, oxidation numbers and disproportionation. **(4)**

(b) Describe electrolytic cells and electrolysis of molten and aqueous electrolytes. **(4)**

Q.4 (a) Explain the standard hydrogen electrode and measurement of E° values. **(4)**

(b) Describe a galvanic cell, salt bridge and photovoltaic cells. **(4)**

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